Countersign QA Report

Phase 1 + Phase 2 first-time QA on Arc testnet. Sign-in works; one approved 0.01 USDC legacy-escrow purchase was signed once and refused by the hub. No funding tx. Wallet still holds 1.0 USDC.

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Verdict

No — not ready for a stranger's first paid attempt today. Sign-in and browse work; the attempted 0.01 USDC purchase signed once and was refused for a deadline mismatch before any funding moved.

A. Verdict

A stranger can open https://web-production-a58de.up.railway.app, read the escrow pitch, browse Registry / Assistant / Seller / Merchant / Start / Verify, complete Privy email OTP (no recovery/passcode path needed in this run), and connect wallet

0xe7ad9A16700b319CA294e059bF7c1FaFf8082AE3 on Arc testnet (chainId 5042002). After login the Buyer phone remains Approval-token gated. With user approval, one exact 0.01 USDC purchase of offer use-arc 0xdd24c0f8…64efc7 was taken through Privy “Sign message” once (EIP-3009 to legacy escrow 0x19c27…1b76, value 10000, validBefore 1793999237 = 2026-11-07 02:37:17 IST). The hub then showed “Signed by your wallet 0xf4d0…971b” and “Refused. The signed deadline is shorter than the offer promises.” No funding transaction, receipt, slip, or refund UI appeared for this job 0x3290a3fc…e29f. Independent RPC re-check after the attempt: USDC balanceOf still 1,000,000 atomic (1.0 USDC @ 6 decimals); no funding logs for that job on the legacy escrow in a 100k-block lookback; authorizationState(address,bytes32) for (wallet, job-as-nonce) returns false/unused. Highest-impact defect: deadline mismatch between prepared authorization and offer promise, with an ~8 minute QA pause for user approval between pre-sign (~02:27 IST) and Sign (~02:35 IST) while the UI still allowed signing the stale prepared auth (no preflight). This does not prove an ordinary fast flow always fails. Studio demos still open without login on the same legacy escrow and show ledger read failures. Net: first impression fails on paid-path reliability and rail/escrow scope clarity even though browse, sign-in, and local verifier checks work.

B. Step / surface table

Step / surface	Result	One-line evidence
1. First 10s impression	Pass*	Clean Incognito used for Phase 1 retest + Phase 2. *Original earliest load was not clean-profile; compact history below.
2a. Landing	Fail	Hero “3 rails live” vs Registry five filters / hub 3 declared vs offers+stats 5 — scope conflict.
2b. Registry	Pass*	Renders sellers/offers; search + rail filter exercised. *Not every control exhaustive.
2c. Offer detail	Pass*	use-arc 0.01 USDC / Arc escrow / 2 min / seller 0x6BE9… shown.
2d. Assistant	Pass*	Purchase card + Hold ≥2s reveals agent command.
2e. Verify (prior slip)	Pass	Prior reachable slip VALID; doubled amount INVALID. This purchase has no receipt/tx — verifier blocked for it.

2f. Seller	Pass*	Seller desk for 0x6BE9… hub-reported earnings (authenticated session).
2g. Buyer	Fail	Still Approval-token gated after login; History/Merchant tabs later showed Sign in (session drop observed, not proven intentional).
2h. Merchant / Start	Pass*	Surfaces render when authenticated; signed-out Merchant showed USDT dashes — do not infer logged-in merchant broken.
3. Sign in	Pass	Privy email OTP succeeded; no recovery/passcode required. Wallet 0xe7ad…2AE3 on Arc testnet.
4. Wallet address/chain	Pass	UI + RPC: 0xe7ad9A16700b319CA294e059bF7c1FaFf8082AE3, chainId 5042002, USDC 0x3600…0000.
5. Purchase to signing	Fail	Signed once; hub refused deadline mismatch; no funding tx. Job 0x3290…e29f unbound on-chain.
6. Refund/deadline vs rail	Fail	Offer 2 min vs signed validBefore window refused by hub; no paid job so no refund path exercised.
7. Receipt + verifier (this buy)	Blocked	No tx hash / slip for this purchase. Prior Phase 1 slip verifier still Pass.
8a. studio.arc.io share	Pass*	Opens without Studio login. Shows legacy escrow + ledger retry error.
8b. Vercel escrow console	Fail	Same legacy escrow (not supplied v2); 0 jobs + “Could not read the latest events”; offer hash differs embed vs direct.
9. Hub readonly rails	Pass	healthz ok commit 872c8cb…; rails escrow/whitechain/hedera-scheduled; offers+stats also hedera/gateway.
10. Funds after attempt	Pass	balanceOf still 1000000 atomic @ 6 decimals at block ~71,089,312 (02:39:28 IST).

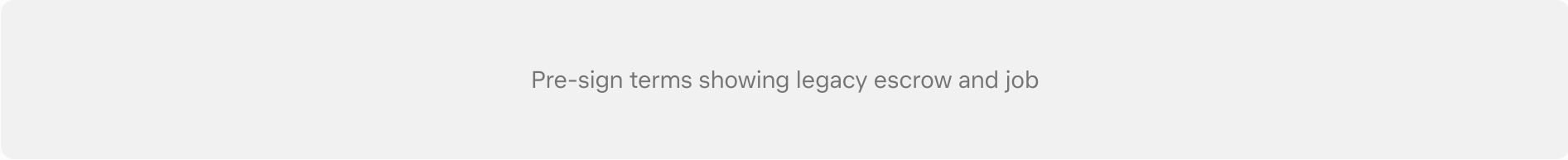
As of 2026-11-07 Asia/Calcutta. Pass = exercised; Fail = defect; Blocked = access/policy or no receipt.

C. Ranked defects

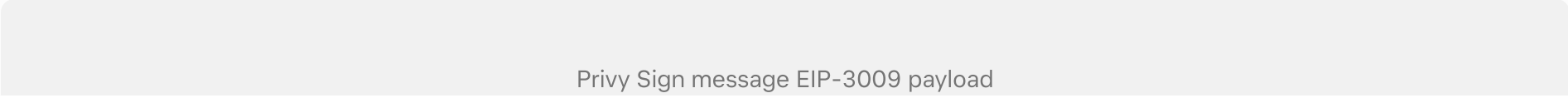
One finding per defect. Ranked by damage to a first paid attempt.

C1. Signed authorization refused: deadline shorter than offer

Severity	● Critical — paid path blocked
URL	https://web-production-a58de.up.railway.app/#/ (Start / buy use-arc)
Repro	Authenticated buyer; open use-arc 0.01 USDC / 2 min / Arc escrow; hold through pre-sign; Sign message once in Privy; observe hub response.
Expected	Prepared EIP-3009 auth matches offer deadline, or UI refuses to open Sign when remaining window is too short.
Actual	UI: “Signed by your wallet 0xf4d0…971b” then “Refused. The signed deadline is shorter than the offer promises.” Typed domain USDC v2 chainId 5042002 verifyingContract 0x3600…0000; message from wallet to legacy escrow 0x19c27…1b76 value 10000 validAfter 0 validBefore 1793999237 (= 2026-11-07 02:37:17 IST) nonce=job 0x3290…e29f. Pre-sign visible ~02:27 IST; Sign ~02:35 IST (~8 min QA pause for user approval). UI allowed signing that prepared auth with no preflight. At Sign, ~2.3 min remained before validBefore. No funding tx/receipt/slip/refund. Treat 0xf4d0…971b as a misleading “Signed by your wallet” identifier (likely signature-hash representation), not proof of a different signer — connected wallet remained 0xe7ad…2AE3. Do not generalize that a fast path always fails.
Console	401 on envelopes/approvals for default 0x0656…ABd8 ≠ session wallet; POST /offers/dd24…/call → 402 after signature; Privy analytics_events CORS/ERR_FAILED.
Fix	Recompute validBefore at Sign time to satisfy offer deadline, or block Sign with an explicit stale-prep warning; align status label with signer address.



Pre-sign — use-arc 0.01 USDC, escrow 0x19c2…1B76, job 0x3290…e29f.



Privy Sign message — will not cost fees; validBefore 1793999237; to legacy escrow.

Hub refused signed deadline shorter than offer

After one Sign — refused deadline mismatch; no funding UI.

C2. Hub/UI use legacy Arc escrow; bonded history is on v2

Severity	High — configuration mismatch
URL	Hub rails + Studio console + explorer
Repro	Compare hub /rails escrow, pre-sign Escrow field, Studio Contract panel, and bonded job logs.
Expected	Live UI escrow matches the escrow that holds current bonded/settlement proofs.
Actual	Live surfaces use legacy 0x19c27D770969291E6A0a305dc1Be1a845E7c1b76. User-supplied Arc escrow v2 0x10A3c07cc6A519C0106e315D2C42010eCbEFF3CE holds the earlier bonded failDelivery job 0xd866a872… (not this refused purchase). Studio Contract panel also shows the legacy address. One finding across hub/UI/Studio — not three.
Console	None required beyond address mismatch.
Fix	Point hub rails + Studio + console pre-sign to the intended escrow deployment and label legacy vs current.

Studio console Contract panel legacy escrow

Studio — Escrow 0x19c27…1b76; Ledger could not read latest events.

Explorer escrow v2 contract

Explorer escrow v2 contract

Explorer — user Arc escrow v2 0x10A3…F3CE (bonded history), not what UI binds.

C3. Rail-scope inconsistency (3 declared vs 5 listed/activity)

Severity	● High — first-screen trust
URL	https://web-production-a58de.up.railway.app/ · #/registry · hub /healthz /rails /offers /stats
Repro	Read hero; Registry filters; GET /healthz, /rails, /offers, /stats.
Expected	One labeled taxonomy for live vs listed vs activity rails.
Actual	Hero “3 rails live”. Hub healthz/rails exact ids: escrow, whitechain, hedera-scheduled (commit 872c8cb4a40d881e94b7eb719bf267803a5592f1, ok:true). Offers also advertise hedera + gateway; /stats.byRail has five keys. Registry shows five filters. Scope conflict / inconsistent messaging — not proof rails are fake or settlements nonexistent.
Console	None for the UI contradiction.
Fix	Align hero, Registry filters, and hub declaration under live vs discovery labels.

Landing hero 3 rails live

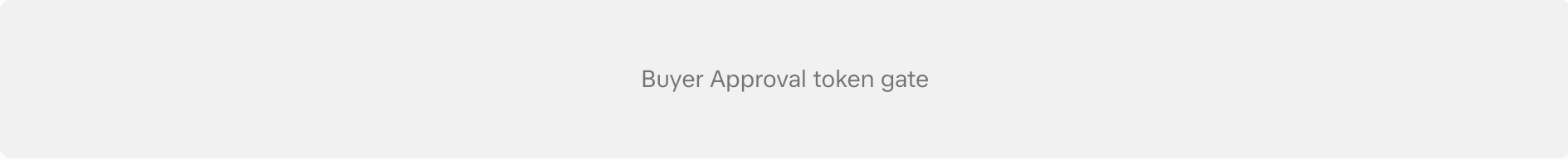
Landing — 3 rails live.

Registry five rail filters

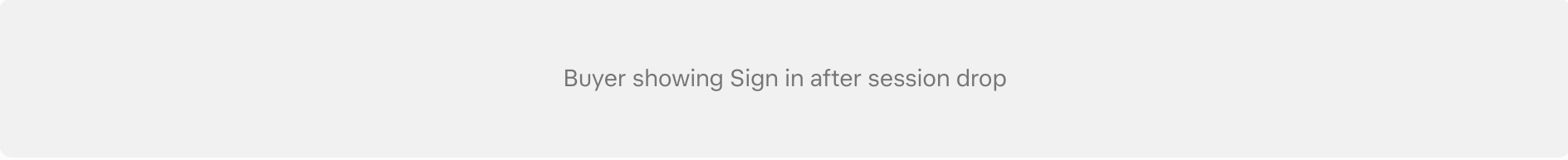
Registry — five rail filters.

C4. Buyer onboarding still token-gated after login

Severity	● High — first-time human path
URL	https://web-production-a58de.up.railway.app/#/buyer
Repro	Sign in; open Buyer; leave Approval token empty.
Expected	Explain envelope + how to obtain a token, or bootstrap after wallet connect.
Actual	Authenticated Phase 1/2 buyer still hits Approval-token wall. Gate not proven broken — acquisition path missing. Later History/Merchant views showed Sign in while the original Start session remained logged in (same Incognito; session persistence bug not proven).
Console	401 envelopes/approvals for default buyer 0x0656… when unauthenticated — expected without token.
Fix	Document/link envelope + token issuance from the empty Buyer state; keep auth cookie consistent across tabs.



Buyer — token required; no acquisition path.

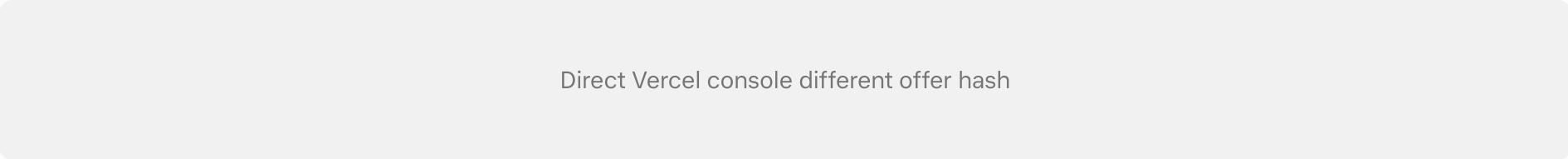


Buyer later — Sign in visible; empty phone (state-limited).

C5. Shared Studio demo: unlabeled sample + ledger read failure

Severity	● High on shared demo URLs
URL	https://studio.arc.io/share/countersign-automated-software-marketplace-9tfb9kfte9 → https://arc-studio-16e0b317ec5f406a8d7d24bf.vercel.app/
Repro	Open share and direct console without Studio login; compare offer hash; watch Ledger.

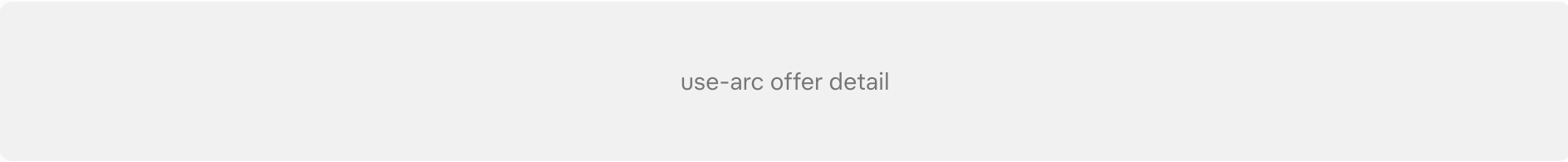
Expected	Labeled demo; stable offer identity; readable ledger or clear offline copy.
Actual	Ready “Extract line items from a PDF invoice” 4.50 USDC, bond 1.00, ~6h countdown, empty seller, Connect wallet. Offer hash differs embed vs direct (locally regenerated draft concern only — not proof of fraud). Ledger: 0 jobs + “Could not read the latest events. Retrying.” Countdown alone is not proof the sample is static/fake. Contract shows legacy escrow + treasury 0xbf9d…fe87 + USDC 0x3600…0000.
Console	Ledger retry copy on-screen; prior runs also saw rpc.testnet.arc.io 429s.
Fix	Label Demo, pin one offer hash, surface last-synced block, point Contract at intended escrow.



Direct console — different offer hash vs share embed; same ledger error.

C6. Unnamed sellers on hub offers

Severity	● Medium — browse trust
URL	Registry / offer pages
Repro	Open Registry or offer detail.
Expected	Display name or explicit anonymity policy.
Actual	“Unnamed seller” + hex. Usability — not proof sellers are malicious.
Console	None.
Fix	Require display name on publish, or state anonymity as policy on every card.



Offer — use-arc (Phase 1 capture).

D. Claims observed that are not fully backed

Scope ambiguity (not proven false)

Hero “3 rails live” vs Registry five filters vs hub three declared ids vs offers/stats five activity keys — inconsistent messaging / scope conflict, not proof of nonexistent rails or fake settlements.

Global “payments that only move when the work lands” is broader than what every listed rail implies unless each settlement model is spelled out.

Unverified / hub-reported only

“Every payment here was real” / live counters — testnet; only sampled settlements chain-checked.

Seller delivery % and refund counts — hub-reported; not recomputed on-chain here.

“43 Foundry tests” — not checked.

Known limit “One blocked payee freezes every payout” — disclosed accepted risk; not independently audited.

Merchant USDT dashes while signed out — do not infer logged-in merchant broken or cross-rail mislabel.

Status “Signed by your wallet 0xf4d0…971b” — treat as misleading identifier label, not unauthorized-signer proof.

Proven matches / withdrawn

Bonded job 0xd866a872… on escrow v2: fund tx 0x55876211… (01:53:36 IST block 71083924) buyer→v2 10000 USDC atomic; close tx 0x973476ed… failDelivery Success (01:53:38 IST block 71083927) DeliveryFailed(jobId,buyer,amount=10000,bond=10000); USDC v2→buyer 0xE062…3f48 20000 atomic = 0.02 USDC. Relayer sender 0xE37F…C267. Non-USDC 18-decimal Transfer in same txs is not a second USDC payout. Matches user’s bond claim; does not match current UI legacy configuration.

This refused purchase created no funding tx — no refund owed; wallet still 1.0 USDC.

Hold control works on true ≥2s press (not permanently broken).

Changing Studio offer hash ≠ proof of a fake published marketplace listing.

E. Missing in the first two minutes

1. Plain explanation of envelope + approval token and how a first-time human gets one after wallet connect.
2. Who can trigger a refund, on which rail, and when (deadline actor — not only “2 min”).
3. One consistent map of live vs listed vs declared rails, and which escrow address is current.
4. Preflight that the authorization validBefore still covers the offer deadline before opening Sign.
5. Seller display names or an explicit anonymity policy.
6. Demo labeling on Arc Studio / Vercel so a ready 4.50 sample is not read as a live listing.

Purchase attempt (Phase 2)

Offer	use-arc 0xdd24c0f8d53f5f4d87630be73cf438eab990ffd63e73f0953e58d4ad3964efc7
Seller	0x6BE98051B852824A3923E3B27f9781Ceb0C0ca4a
Amount	0.01 USDC (10000 atomic)
Deadline shown	2 min
Escrow in UI	legacy 0x19c27d770969291e6a0a305dc1be1a845e7c1b76
Job	0x3290a3fc72af261420fd071167f78bdc651e490719e750b89bcab2614047e29f
Signatures	Exactly one Privy Sign and continue — no retries, no token approval tx
Hub result	Refused — signed deadline shorter than the offer promises
Funding tx	None (not available — do not fabricate)
validBefore	1793999237 → 2026-11-07 02:37:17 IST (Asia/Calcutta). Signature existed; at Sign (~02:35) still before expiry; by independent verify (~02:39) epoch had passed. No cancellation promised or performed.

Timing honesty

Pre-sign context was visible from ~02:27 IST; actual Sign ~02:35 IST after an ~8 minute pause for user approval of the exact 0.01 spend. The UI still offered Sign on that prepared authorization. This QA does not claim an ordinary uninterrupted fast checkout always fails the same way.

Independent chain verification

RPC	https://rpc.testnet.arc.io (read-only eth_call / eth_getLogs)
chainId	5042002 (0x4cef52)
USDC	0x3600000000000000000000000000000000000000 — decimals 6 verified
Wallet	0xe7ad9A16700b319CA294e059bF7c1FaFf8082AE3
balanceOf	1000000 atomic = 1.0 USDC at block 71089312 (2026-11-07 02:39:28 IST)
Job funding on legacy	0 logs for job 0x3290…e29f on 0x19c27…1b76 over 100k-block lookback; 0 USDC Transfer wallet→legacy over 50k blocks
authorizationSta…	EIP-3009 authorizationState(address,bytes32) selector 0xe94a0102 → 0 (unused/not invalidated) for (wallet, job nonce). uint256 overload reverts. Signature bytes not read.

Explorer failDelivery success 0.02 USDC return on v2

Prior bonded close — failDelivery Success; 0.02 USDC from CountersignedEscrow to 0xE062… (v2 history, not this purchase).

Full screen audit

Landing / Registry / Offer / Assistant

Landing claims escrow-held agent payments and “3 rails live”. Registry lists five rail filters and searchable offers. use-arc offer

shows 0.01 USDC, Arc·escrow, 2 min. Assistant Hold ≥2s reveals agent CLI. Signing path covered in Purchase section.

Registry listing

Registry listing.

Hold success reveals agent command

Hold success — agent command revealed.

Buyer / Merchant

Authenticated buyer remains token-gated. Additional tabs later showed Sign in while Start remained connected — observed, not proven as intended product behavior. Signed-out Merchant USDT dashes are state-limited evidence only.

Merchant signed-out USDT dashes

Merchant while signed out — USDT dashes; do not over-infer.

Sign-in

Phase 2: Privy email OTP succeeded; wallet connected on Arc testnet; no recovery or passcode step required in this run. (Phase 1 compact history: earlier blocked before OTP when no mailbox was available.)

Verifier prints

This purchase

No transaction hash / slip for job 0x3290…e29f — verifier for THIS purchase is blocked. Prints below are from the prior reachable Phase 1 slip (not this attempt).

Prior verified slip (full verify URL)
web-production-a58de.up.railway.app

Slip / verify id	0xff8976e41eee194a0294be88d394e356f20dbfe243386b3c5b1b9212ee8eb2a9
Transaction	0x3ce0d4341797435ded37b5ad6e9b437404e590033def483b3dd902dc65a67e1e
Seller	0x6BE98051B852824A3923E3B27f9781Ceb0C0ca4a
network / chainId	eip155:5042002 (Arc testnet)
rail	escrow
amount (valid body)	10000

Before chain — observed UI (prior slip)

> VALID. Hash and signatures checked in this browser. Nothing was sent anywhere and the chain was not asked.

The hash commits to every field.

Keccak-256 of the body in canonical JSON is the slip hash, so no field has changed.

Signed by the seller.

EIP-712 Release(jobId, slipHash), under the escrow contract’s own domain.

The buyer agreed to this offer, payee and amount.

The countersigned terms derive this slip’s job id.

Countersigned by the buyer.

One EIP-3009 signature names the escrow and binds the amount and the job.

Settlement not read from the chain.

The slip names the transaction below. Read it from Arc testnet here, or open it on the explorer.

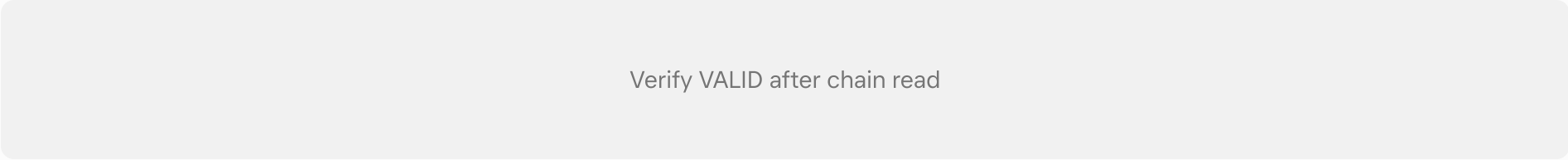
After Check the chain — observed UI (prior slip)

> VALID. Hash and signatures checked in this browser. The release was read from Arc testnet’s public RPC.

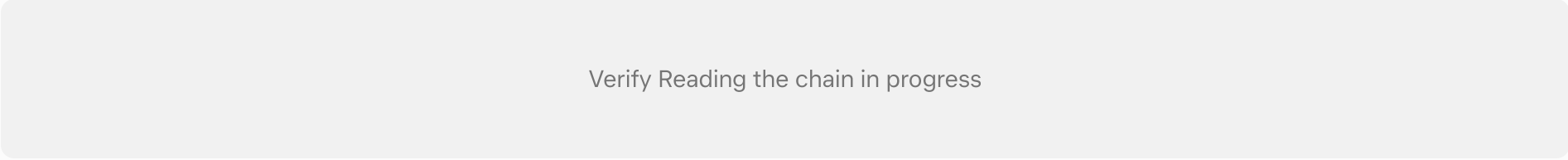
Known escrow: Arc testnet, deployed through Arc Studio.

The chain holds this slip’s hash.

Arc testnet block 71,069,395: the escrow’s Released event for this job carries exactly the hash computed here.



Prior slip — VALID after chain.



Capture during Reading the chain (in progress).

Tampered amount — observed UI (prior slip)

Helper: same slip with amount doubled after signing. Amount 10000 → 20000.

> INVALID
>
> Body hashes to 0x1782…b444, slip says 0xff89…b2a9.
>
> Signed by 0x92c5…064F, not the seller 0x6BE9…ca4a.
>
> The countersigned terms differ from the slip.

Verify INVALID doubled amount

INVALID — doubled amount helper.

External demos

1. Arc Studio share — Countersign Automated Software Marketplace
   studio.arc.io
2. Countersign escrow console (Vercel)
   arc-studio-16e0b317ec5f406a8d7d24bf.vercel.app

studio.arc.io share · Opens / flawed

Opens without Studio login. Community disclaimer. Open app → Vercel. Legacy escrow on Contract panel; ledger retry error.

Vercel escrow console · Opens / flawed

Opens without Studio login. Ready 4.50 sample, empty seller, differing offer hash vs share, ledger read failure, legacy escrow (not v2).

Prior Phase 1 history (compact)

Early Phase 1 (before mailbox): sign-in blocked at Privy before OTP; no wallet observed; purchase signing not reached; clean first-ten-seconds on the very first load was not met (Incognito used afterward). Those stops are superseded by Phase 2 outcomes above (OTP succeeded; wallet connected; one Sign refused). Offline verifier disconnect was not conducted in either phase.

Evidence appendix

- [x] ~~Testnet only — no mainnet funds~~